

Virtual reality in Angkor's heritage trust

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ABSTRACT

This reflection report critically evaluates the strategic feasibility of implementing virtual reality (VR) technology for Angkor Heritage Trust (hereinafter referred to as "Trust"). The Trust is a medium-sized organization responsible for the protection and promotion of Angkor Archaeological Park in Cambodia. The main challenge it faces is defined as reducing the serious physical degradation caused by over-tourism, while satisfying the global tourists' desire for deep and intimate participation. The proposed solution is the "Lost Angkor" immersive VR center located outside the park, which aims to rebuild the original color and architectural glory of the temple. The research results show that VR provides a powerful and non-invasive method to achieve protection goals and improve learning outcomes, especially by cultivating tourists' pro-sustainable behavior. However, its successful deployment depends on overcoming major ethical obstacles about authenticity and solving major operational defects related to large-capacity passenger flow and sanitation in tropical climate. This report suggests that pilot investment should be made in stages, giving priority to historical accuracy and stable and customized hardware.

1. INTRODUCTION

The reason for choosing this case lies in an enlightening personal experience. When I experienced the version of Google Earth VR in this course, I had the opportunity to "visit" Angkor Wat. This prompted me to think: If a non-immersive VR experience can be so shocking, then, can a well-designed VR solution become a necessary and extensible tool to deal with the double dilemma faced by Angkor Wat (that is, the conflict between protection and public access)? This report focuses on this issue.

The purpose of this study is threefold: to define the key protection and access issues faced by the trust; Put forward a concrete and high-value VR intervention measure-"The Lost Angkor Center"; The practical, ethical and operational limitations of this solution in the context of sensitive heritage are discussed rigorously and critically. The core assumption is that VR can be used as a dual-purpose tool, serving both the protection mission and the education mission, provided that it actively manages its inherent technical and philosophical risks.

2. BACKGROUND AND PROBLEM CONTEXT

2.1 The Client and Environmental Fragility

Angkor Heritage Trust manages a huge temple complex, and its operation follows the principle that sustainable protection takes precedence over commercial income. Basic materials-sandstone and laterite-are easily affected by environmental pressure. Continued

physical foot wear and structural pressure caused by crowding are direct threats. The high humidity accumulated by human breath in closed areas (such as inner sanctuaries and narrow corridors) accelerates the decay of rocks and any remaining murals. Physical barriers need to be set up for continuous human activities. Although these barriers are essential for protection, they limit the activities of tourists.

2.2 Defining the Dual Conflict

The trust management dilemma can be divided into two conflicts. The first is the cognitive gap and the weakening of authenticity. Visitors are restricted to watch the monument from a distance and can only see the gray and weathered ruins. Physically, they are forbidden to observe complicated details, and psychologically, they can't completely rebuild the original glory of the complex in the 12th century-vivid colors, gilded surfaces and comprehensive water conservancy system. This limited perspective leads to a feeling of missing opportunities and weakens tourists' perception of the authenticity of existence. This is the subjective emotional connection with the past of the site, which has been proved to be very important for tourists' satisfaction. The second one is the conflict of physical intervention. Here, the demand of absolute physical non-intervention (no touching and approaching) directly conflicts with the demand of global tourists for high participation and details. Current measures, such as protecting railings and security personnel, manage risks, but ultimately fail to meet the full scope of the educational mission. Therefore, the strategic goal of trust is quantitative and qualitative: Using immersion technology to provide a richer, more detailed and non-physical educational experience, this will reduce the number of tourists in the most vulnerable core areas and reduce the environmental impact.

3. VR AFFORDANCES AND PROPOSED SOLUTION

3.1 VR as a Preservation Technology

Different from augmented reality (AR), AR is to superimpose digital information into the real world, and the power of VR lies in its "complete substitution" ability. This isolation makes it a clear solution for heritage sites that need absolute non-contact protection. There are three core availability related to the trust. At first, zero-risk access was established, because VR created a perfectly controllable environment, eliminating all threats caused by human flow, humidity and direct physical contact. Secondly, the reconstruction of time and space allows VR to reconstruct the lost or damaged heritage, which provides the only way to visualize the original state of the building complex, this is the key factor to enhance cognitive participation and historical empathy. Finally, by cultivating deep emotional and cognitive connections, behavior guidance is realized. Make VR as a guiding tool, positively influence the attitude of tourists, and lead them to produce measurable pro-sustainable behavior after entering the actual heritage site.

3.2 Proposal: The "Lost Angkor" Immersive Center

The proposed solution is to develop the "Lost Angkor" immersive VR center outside the main tourist center in Siem Reap.

Concept and characteristics: The center will be a high-throughput theater, using 80 fixed seat units with synchronous motion and equipped with commercial VR head display. This 15-minute project aims to attract the public and achieve rapid turnover, and will include a

sequential journey covering three core parts. This includes color rebirth, that is, close-up and high-fidelity reconstruction of bayon temple and Angkor Wat, restoring the original red, blue and gold tones, and allowing tourists to "touch" the relief visually. Next is a visit to the inner sanctuary, which simulates entering the inner chamber with fragile or restricted structure and watching the ancient architectural techniques and ritual practices. The last part is the Khmer Water Conservancy Empire, which flies over the rebuilt city and its vast and well-functioning water management system to provide key background knowledge.

Impact: By satisfying the desire for intimacy and detail in a controlled environment, it has achieved its main goal of diverting tourists from the most sensitive core temples.

4. CRITICAL ANALYSIS AND DISCUSSION

The implementation of the "Lost Angkor" VR initiative has brought great challenges. Strict scrutiny is needed to ensure that investments serve protection.

4.1 The Authenticity and Ethical Challenge

The most profound limitation of VR proposal lies in the paradox of authenticity-that is, between physical relics and users' subjective experience. VR content must be a historical interpretation. Especially, the reconstruction of the original color scheme needs academic speculation. Trust is faced with ethical responsibility, and it is necessary to distinguish between verifiable facts and artistic permission. If the reconstructed virtual environment is considered to be more real or beautiful than the real temple, this may eventually devalue the physical heritage in the eyes of tourists.

4.2 Operational and Technical Limitations

High-capacity tourism brings practical friction that VR technology is often difficult to cope with. First of all, regarding hygiene and throughput, disinfection among users is a major logistics and cost obstacle in a high-flow and tropical environment. Failure to maintain strict hygiene regulations in VR center will significantly affect user satisfaction and public health. This necessity imposes a hard limit on the tourist throughput of facilities, which may cause bottlenecks, thus undermining the goal of reducing congestion. Secondly, there are problems in the stability of content and technology, because it is very expensive to build a high-fidelity 3D model of the whole Angkor Wat. In addition, motion sickness is still a common technical risk. In order to minimize the chance of motion sickness and maximize operational efficiency, the trust must strive to adopt a fixed-seat, low-interaction design, sacrificing some potential of VR for comprehensive exploration to ensure reliability.

4.3 Connection to Sustainable Goals

Despite the challenges, VR solutions strongly support the trust's sustainable goals. Because the VR experience is a powerful prelude to physical visits by enhancing cognitive understanding and emotional awe. By deepening tourists' sense of connection and value to the fragility of the site, the trust uses technology to promote management.

5. CONCLUSION

The strategic implementation of the "Lost Angkor" VR Center provides a feasible and powerful solution for Angkor Heritage Trust to cope with the conflict between its heritage protection and the participation of global tourists. This technology provides a unique opportunity to enhance the educational experience. While actively slowing down physical degradation. The deployment of trust is suggested to be constructed in order. Crucially, VR hardware that is highly durable and easy to disinfect is needed. Trust should abide by the high-capacity operation agreement to minimize the friction points such as hygiene and solve practical problems in high-throughput public environment.

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